Yizhi Liang

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SUMMARY

PhD student in Health Economics at USC with industry HEOR experience in both pharmaceutical and consulting companies and multiple first-author peer-reviewed publications in health economics, health policy, and epidemiology. Applied depth in health-economic modeling (cost-effectiveness, cost-utility, and budget-impact models, value-of-information analysis), real-world evidence and causal inference, HTA and reimbursement pathways, and AI integration in HEOR applications, with disease-area work in oncology, infectious disease, cardiometabolic and chronic disease, and rare pediatric conditions.

EDUCATION

Alfred E. Mann School of Pharmacy and Pharmaceutical Sciences, University of Southern California Aug 2024 – Present
PhD in Health Economics (GPA: 3.92/4.00) Los Angeles, CA USA

Harvard T.H. Chan School of Public Health, Harvard University Aug 2022 – May 2024
MS in Global Health and Population (GPA: 3.84/4.00) Boston, MA USA
Thesis: Evolving Trends in Racial and Ethnic Disparities in Syphilis Prevalence Among Pregnant Women in the United States from 2014 to 2022.

West China School of Public Health, Sichuan University Sep 2017 – Jun 2022
MBBS in Public Health (GPA: 3.71/4.00) · ***BMgmt in Financial Management*** Chengdu, Sichuan China
Thesis: Study on the Association Between Quality of Outpatient Care and Patient Satisfaction in Rural China: A Cross-Sectional Standardized Patient Study.

SKILLS

- **Health Economic Modeling:** Cost-effectiveness and cost-utility analysis, Markov cohort models, partitioned survival models, decision-tree models, microsimulation models, budget impact analysis, extended cost-effectiveness analysis, deterministic and probabilistic sensitivity analysis, value-of-information analysis
- **AI-Integrated HEOR:** AI-assisted quality control of health economic models, large language model (LLM) review workflow and evaluation design, benchmark construction, systematic evaluation of LLM review performance, human-in-the-loop process design
- **AI System Design and Reliability:** Agentic system design across the tool, context, and prompt layers, context engineering, prompt engineering, failure-mode analysis for production AI systems, silent-failure detection and explicit unable-to-verify escalation to human reviewers, prompt-injection mitigation (untrusted model inputs treated as data, not instructions)
- **Machine Learning and Statistical Modeling:** Supervised machine learning, nested resampling and cross-validation, Bayesian hyperparameter optimization, algorithmic fairness evaluation, Bayesian mathematical modeling
- **Causal Inference and Real-World Evidence:** Difference-in-differences, fixed-effects panel models, propensity score matching, marginal structural models, complex survey data analysis (MEPS, NHANES), quantile regression, quantitative bias analysis
- **HTA and Evidence Review:** Health technology assessment and reimbursement pathways, Delphi panel methods, systematic and scoping reviews, model verification, validation, and calibration
- **Programming and Tools:** R, Python, Stata, SAS, SQL, Stan, Julia, TreeAge, Excel-based health economic modeling, Git, reproducible analysis pipelines
- **Disease Areas:** Infectious disease (syphilis, congenital syphilis, HIV and PrEP), oncology, cardiometabolic and chronic disease, pediatric neuroimmune conditions (PANS/PANDAS), dermatology, chronic respiratory disease

PROFESSIONAL EXPERIENCE

Merck
Health Economics Intern, 2026 Future Talent Program Jun 2026 – Aug 2026

- Built an AI-assisted quality-control (QC) workflow for Excel-based health economic models, including cost-effectiveness and budget-impact models with Markov, partitioned-survival, and decision-tree structures.
- Translated hands-on modeling and QC experience into AI-executable review checks covering the full model lifecycle, from structure and parameter sourcing through the calculation engine, base-case results, and uncertainty analysis to transparency and documentation, prioritized by the failure modes that most often break economic models.
- Established a reviewer-facing standard operating procedure built on an evidence-backed, source-traced, multi-agent review framework, with a human-in-the-loop design that reserves scoping, finding adjudication, and final accountability for the reviewer, requires an explicit unable-to-verify signal in place of a silent pass whenever a check exceeds AI capability, and treats all workbook content as data under review rather than instructions to be followed.
- Designed and executed a systematic performance evaluation of the workflow, specifying the evaluation plan and metrics, assembling an evaluation benchmark from existing model assets, and building a health-economic-model knowledge base to support the review checks.

Stratevi
Summer Intern Jun 2025 – Aug 2025

- Performed quality control (QC) of Excel-based economic models from a U.S. payer perspective, including a cost-effectiveness model for a surgical dermatology intervention as well as disease burden and early cost-effectiveness models in cardiovascular disease.
- Conducted real-world data analysis in R for an oncology targeted therapy, constructing the study cohort and deriving treatment patterns, adherence, discontinuation reasons, and time-to-event outcomes for the study report.
- Reviewed the statistical analysis plan for a longitudinal real-world study in chronic respiratory disease, drafted a real-world evidence proposal using propensity score matching for indirect treatment comparison, and delivered a targeted literature review on unmet need and disease burden in stroke prevention in atrial fibrillation.
- Developed internal market access reference materials on the German AMNOG pathway, covering the healthcare system, regulatory and health technology assessment procedures, and pricing and reimbursement mechanisms, and contributed to an Anatomical Therapeutic Chemical (ATC) classification submission to the World Health Organization through precedent analysis, dossier drafting, and quality assurance.
- Drafted internal best-practice guidance for designing and implementing Delphi panel studies, and built and maintained internal R and Python repositories of open-source cost-effectiveness models with data and code sources integrated and executability verified.

Harvard T.H. Chan School of Public Health
Teaching Assistant Jan 2024 – May 2024

- Led course labs for GHP 501 Modeling for Health System Analysis & Priority Setting and GHP 201 Advanced Modeling for Health System Analysis & Priority Setting.
- Taught R for data analysis and economic evaluation, with emphasis on cost-effectiveness analysis and extended cost-effectiveness analysis.

Dept. for HIV/AIDS Prevention and Control, Zhuhai Center for Disease Control and Prevention
Intern Aug 2021 – Nov 2021

- Built an automated system in R and Stata to generate local monthly HIV/AIDS epidemic reports from surveillance data.

- Supported HIV/AIDS research on social-network-based HIV self-testing promotion among men who have sex with men.

West China Hospital of Sichuan University
Medical Intern Aug 2020 – Dec 2020

- Performed daily screening and diagnosis of common diseases.
- Delivered patient health education as part of clinical rotations.

RESEARCH EXPERIENCE

Leonard D. Schaeffer Center for Health Policy & Economics, University of Southern California
Research Assistant

Project: Cost-Effectiveness and Budget Impact of IVIG versus Corticosteroids for Moderate-to-Severe Pediatric PANS/PANDAS — Jan 2026 – Present

- Developed a Markov cohort model in R estimating ICERs and net monetary benefit for IVIG versus corticosteroids alone from Medicaid and societal perspectives, with probabilistic sensitivity and budget impact analyses.
- Built a renv-locked, fully reproducible analysis pipeline with automated validation.
- Manuscript in preparation targeting Value in Health.

Project: Waiting for Trial Evidence or Using Real-World Evidence: Trade-Offs for Coverage with Evidence Development — Dec 2025 – Oct 2026

- Derived the policy-indifference bias threshold at which real-world-evidence-based coverage with evidence development equals RCT-based coverage in population expected net benefit.
- Characterized the threshold using quantitative bias analysis and sensitivity analyses on timing advantage and trial cost in Python simulation.
- Published in PharmacoEconomics (2026).

Project: The Differential Impact of Family-Level Multimorbidity on Financial Hardship by Insurance Status: Evidence from MEPS 2018–2023 — Aug 2024 – Present

- Analyzed the 2018–2023 Medical Expenditure Panel Survey (about 67,000 family-years) to estimate how family-level multimorbidity relates to financial hardship, including high out-of-pocket burden, medical debt, and cost-related barriers to care, across private, public, and uninsured coverage.
- Built the primary analysis pipeline in Stata with R for publication-ready tables and figures, using survey-weighted probability models with year and panel fixed effects.
- Manuscript under revision.

Project: Evaluating Policy Options for New Treatments Under Uncertain External Evidence — Aug 2024 – Apr 2026

- Designed a novel conceptual diagram to compare policy options on new treatment with external evidence, from never granted without a local trial, delayed access pending a local trial, immediate access without a local trial, and immediate access while awaiting a local trial.
- Developed a partitioned survival model to estimate the monetary values from the combination of sintilimab and chemotherapy or the chemotherapy only for treating first-line locally advanced or metastatic nonsquamous non-small-cell lung cancer to illustrate the conceptual diagram.
- Employed Python to compute the expected value of perfect information (EVPI), the expected value of sample information (EVSI), and the expected net benefit of sampling (ENBS) using a two-level double-loop Monte Carlo approach, comparing the expected monetary benefits of four policy options.
- Published in Value in Health (2026).

Harvard T.H. Chan School of Public Health
Graduate Research Assistant

Project: Evaluating Fairness Across Machine Learning Algorithms in Health Models Incorporating Race/Ethnicity as Predictors — Aug 2023 – May 2024

- Employed Python to build machine learning models to predict the risk of cardiovascular disease in U.S. Adults based on NHANES data and the risk of adverse pregnancy outcomes through U.S. live birth certificate data.
- Deployed the nested resampling and cross-validation approach based on OPTUNA Bayesian framework to optimize hyperparameters for prediction models and ensure generalizability.
- Examined the racial and ethnic differences in model accuracy under different scenarios, such as including race/ethnicity as predictors, building race/ethnicity-specific models, preprocessing to balance race/ethnicity observations, and post-processing predictions.

Project: Evolving Trends in Racial and Ethnic Disparities in Syphilis Prevalence Among Pregnant Women in the United States — Sep 2023 – May 2024

- Built a Bayesian mathematical model in R to estimate trends in the prevalence of syphilis by race/ethnicity among pregnant women in the US between 2016 and 2023.
- Investigated the racial/ethnic disparities in the prevalence of syphilis among pregnant women over the study period, quantified by a set of disparity metrics.

Project: Estimating PrEP Need Among People Who Inject Drugs (PWID) in the US: An Incidence Estimation Study — May 2023 – May 2024

- Constructed a Bernoulli model in R to estimate state-level HIV risk among needle-sharing PWID in the U.S. with the multivariate sensitivity analyses conducted.

Project: Divergent Impacts on Health and Economic Equity: The Community-Based Hypertension Care Program in Ghana — Jan 2023 – May 2024

- Developed a microsimulation model in R using the extended cost-effectiveness analysis framework to examine the distributional impact of a community-based hypertension management program in Ghana.
- Simulated distributional health outcomes and financial outcomes by wealth quintiles with a concentration index to reveal health equity and economic equity to assist in multi-criteria decision-making.

Project: Inequalities in Access and Quality of Prenatal Care to Racial/Ethnic Disparities in Congenital Syphilis Burden — Nov 2022 – May 2024

- Conducted scoping literature reviews to parameterize the mathematical model used in this project.
- Led the data analysis in R on birth file data in the U.S., including data cleaning and computing prior distributions on key parameters, using bootstrapping techniques to obtain parameter values for categorical variables.

Rural Education Action Program, Stanford Center on China's Economy and Institutions
Research Assistant

Project: Telemedicine Service Promotion in Rural China — Aug 2019 – Aug 2022

- Conducted a baseline survey of 167 rural townships enrolled in a telemedicine program.
- Developed questionnaires, trained investigators, and ensured data quality.
- Assessed online healthcare quality offered by Direct-to-Consumer telemedicine platforms in China.
- Conducted the difference-in-differences method in Stata to draft the evaluation report of the program.

Project: Quality of Primary Care in Rural China — Sep 2019 – Jun 2022

- Used the standardized patient method and the clinic vignettes to assess the quality of primary care in the field.

- Employed Stata to describe the relationship between outpatient satisfaction and received medical quality using quantile regression.

West China School of Public Health, Sichuan University
Research Assistant

Project: An Investigation on the Quality of Care in Rural China — Aug 2019 – Jun 2022

- Contributed to a China Medical Board (CMB) project researching the quality of care for non-communicable diseases (NCDs) in rural China.
- Wrote an undergraduate thesis on the study of satisfaction with rural health services.

Project: Southwest Regional Natural Population Cohort Study — Mar 2021 – Aug 2021

- Used propensity score method and marginal structural model to explore the association between passive smoking exposure and insomnia in a plateau Tibetan population.

Peking Union Medical College and Chinese Academy of Medical Sciences
Research Assistant

Project: Vaccine Delivery Quality Improvement Interventions in Elementary Schools — Aug 2020 – Sep 2020

- Conducted literature review about the social and behavioral theories on vaccine hesitancy.
- Co-wrote main sections in the Project Application for the CMB project that evaluated the cost-effectiveness analysis of Flu vaccine delivery in Beijing.

PUBLICATIONS

1. Chen J*, **Liang Y***, Hu M, Verguet S, Jiao B. Catastrophic Health Expenditure Underestimates Financial Hardship in Chronic Disease Households. Am J Manag Care. In press. (*Equal contribution.)
2. Jiao B, **Liang Y**, Tao L, Zafari Z. Waiting for Trial Evidence or Using Real-World Evidence: Trade-Offs for Coverage with Evidence Development. Pharmacoeconomics. 2026 Oct 9. doi: 10.1007/s40273-026-01668-3. Online ahead of print. PMID: 42853373.
3. **Liang Y**, Jiao B. Evaluating Policy Options for New Treatments Under Uncertain External Evidence. Value Health. 2026 Sep;29(9):1618-1625. doi: 10.1016/j.jval.2026.03.2240. PMID: 42000135.
4. Rönn MM, Kourtis AP, **Liang Y**, Zheng L, Puente T, Huang YA, Zhu W, Patel RR, Wiener J, Hoover KW, Van Handel M, Menzies NA, Salomon JA. Quantifying the number of people who would benefit from HIV pre-exposure prophylaxis (PrEP) in the United States: a comparison of behavioral, acquisition-risk based, and economic metrics. Ann Epidemiol. 2026;118:110079. doi: 10.1016/j.annepidem.2026.110079. PMID: 41856300.
5. **Liang Y**, Menzies NA, Rönn MM. Trends in syphilis prevalence by race and ethnicity among people who are pregnant in the United States 2016-2023. Sex Transm Dis. 2025;52(9):535-541. doi: 10.1097/OLQ.0000000000002187. PMID: 40417976.
6. Rao M, Chen SY, **Liang Y**, Yu J. Underdiagnosis of Allergic Contact Dermatitis in Under-Represented Groups: An All of Us Database Analysis. Dermatitis. 2025 Mar-Apr;36(2):170-172. doi: 10.1089/derm.2023.0263. Epub 2023 Dec 29. PMID: 38156811.
7. Wu Y, **Liang Y**, Cai Z, Li L, Sun C, Sylvia S, Zhou H, Feng J, Rozelle S. Process quality, diagnosis quality, and patient satisfaction of primary care in Rural Western China: A study using standardized patients. Patient Educ Couns. 2024 Jun;123:108208. doi: 10.1016/j.pec.2024.108208. Epub 2024 Feb 15. PMID: 38377708.
8. **Liang Y**, Lin Y, Jiao B. Health Interventions May Have Divergent Impacts on Health and Economic Equity: A Case Study of the Community-Based Hypertension Improvement Project in Ghana. Appl Health Econ Health Policy. 2024 May;22(3):353-362. doi: 10.1007/s40258-024-00871-7. Epub 2024 Jan 24. PMID: 38267804.

9. Rao M, Chen SY, **Liang Y**, Yu J. Underdiagnosis of psoriasis in underrepresented groups: An “All of Us” database analysis. J Am Acad Dermatol. 2023 Dec;89(6):1279-1282. doi: 10.1016/j.jaad.2023.08.011. Epub 2023 Aug 12. PMID: 37579839.
10. Du Y, Zhou Q, Cheng W, Zhang Z, Hoelzer S, **Liang Y**, Xue H, Ma X, Sylvia S, Tian J, Tang W. Factors Influencing Adoption and Use of Telemedicine Services in Rural Areas of China: Mixed Methods Study. JMIR Public Health Surveill. 2022 Dec 23;8(12):e40771. doi: 10.2196/40771. PMID: 36563026; PMCID: PMC9823570.
11. Cheng W, Zhang Z, Hoelzer S, Tang W, **Liang Y**, Du Y, Xue H, Zhou Q, Yip W, Ma X, Tian J, Sylvia S. Evaluation of a village-based digital health kiosks program: A protocol for a cluster randomized clinical trial. Digit Health. 2022 Oct 5;8:20552076221129100. doi: 10.1177/20552076221129100. PMID: 36211797; PMCID: PMC9537487.
12. He R, **Liang Y**, Lin C, Guo B. Association between Passive Smoking Exposure and Insomnia in a Tibetan Non-Smoking Population Based on the Propensity Score. Chinese Journal of Disease Control & Prevention. 2022;26(7):815-819. doi: 10.16462/j.cnki.zhjbkz.2022.07.012.
13. Lyu H, Tang H, **Liang Y**, Huang S, Wang Y, Huang W, Zhou Y. Alcohol Consumption and Risk of Liver Fibrosis in People Living With HIV: A Systematic Review and Meta-Analysis. Front Immunol. 2022 Mar 18;13:841314. doi: 10.3389/fimmu.2022.841314. PMID: 35371091; PMCID: PMC8971654.

PREPRINTS

1. Rönn MM, **Liang Y**, Bronsard M, et al. Racial and ethnic differences in congenital syphilis: mathematical modeling study analyzing the role of prenatal care. medRxiv 2026.01.30.26345236. doi: 10.64898/2026.01.30.26345236.

CONFERENCE PRESENTATIONS

1. Liang, Y., & Jiao, B. (2026). The Differential Impact of Family-Level Multimorbidity on Financial Hardship by Insurance Status: Evidence from MEPS 2018–2023. *ISPOR 2026*, May 17–20, 2026, Philadelphia, PA. Value in Health, 29(S6), Abstract HPR68.
2. Chen, J.*, Liang, Y.*, Hu, M., Verguet, S., & Jiao, B. (2026). Catastrophic Health Expenditure Substantially Underestimates Financial Hardship in Households with Chronic Conditions. *ISPOR 2026*, May 17–20, 2026, Philadelphia, PA. Value in Health, 29(S6), Abstract HPR153. (*Equal contribution.)
3. Liang, Y., Sheng, C., Luo, J., & Chen, B. (2025). Evaluating Fairness Across Machine Learning Algorithms in Health Models Incorporating Race/Ethnicity as Predictors. *ISPOR 2025*, May 13–16, 2025, Montréal, QC, Canada. Value in Health, 28(S1), Abstract P14.
4. Liang, Y., & Jiao, B. (2025). Private Insurance May Fail to Protect Against Healthcare Financial Hardship: Evidence from the 2018 and 2019 Medical Expenditure Panel Survey. *ISPOR 2025*, May 13–16, 2025, Montréal, QC, Canada. Value in Health, 28(S1), Abstract HPR170.
5. Liang, Y., Menzies, N., & Rönn, M. (2025). Evolving Trends in Racial and Ethnic Disparities in Syphilis Prevalence Among People Who Are Pregnant in the United States, 2016–2023. *ISPOR 2025*, May 13–16, 2025, Montréal, QC, Canada. Value in Health, 28(S1), Abstract PT30.
6. Liang, Y., & Jiao, B. (2025). Leveraging External Clinical Evidence vs. Awaiting Local Confirmatory Trial: A Value of Information Approach. *The 2025 Western PharmacoEconomics (WPEC) Conference*, March 31 – April 2, 2025.
7. Liang, Y., & Jiao, B. (2025). Private Insurance May Fail to Protect Against Healthcare Financial Hardship: Evidence from the 2018 and 2019 Medical Expenditure Panel Survey. *The 2025 Western PharmacoEconomics (WPEC) Conference*, March 31 – April 2, 2025.
8. Liang, Y., Menzies, N., & Rönn, M. (2025). Evolving Trends in Racial and Ethnic Disparities in Syphilis Prevalence Among People Who Are Pregnant in the United States, 2016–2023. *The 2025 Western PharmacoEconomics (WPEC) Conference*, March 31 – April 2, 2025.
9. Rönn, M., Liang, Y., Bronsard, M., Gift, T., Myles, R., Chesson, H., Menzies, N., Sabety, A., Hsu, K., & Salomon, J. (2024). Contribution of Inequalities in Access and Quality of Prenatal Care to Congenital Syphilis Burden. *The 2024 STI Prevention Conference*, September 16–19, 2024, Atlanta, GA.

10. Liang, Y., Sheng, C., & Chen, S. Y. (2024). Racial and Ethnic Fairness in Risk Prediction Models for Cardiovascular Diseases Among US Adults. *ISPOR 2024*. Value in Health, 27(6), S3.
11. Chen, P., & Liang, Y. (2024). Optimization in Universal Health Coverage Design in South Africa Considering Both Health Outcomes and Health-Induced Poverty Outcomes. *ISPOR 2024*. Value in Health, 27(6), S219.
12. Liang, Y., Lin, Y., & Jiao, B. (2023). Divergent Impacts on Health and Economic Equity: The Community-Based Hypertension Care Program in Ghana. *The 45th Annual North American Meeting of the Society for Medical Decision Making*, October 22–25, 2023.

AWARDS & HONORS

Outstanding Research Assistant, USC Alfred E. Mann School of Pharmacy and Pharmaceutical Sciences	May 2026
Best Student Poster Research Presentation, ISPOR 2025	May 2025
Top 5% Finalist for the ISPOR 2025 Research Presentation Awards	Mar 2025
Top 5% Finalist for the ISPOR 2024 Research Presentation Awards	Mar 2024
Lee B. Lusted Student Prize Finalist, The 45th Annual Meeting of the Society for Medical Decision Making	Oct 2023
Harvard T.H. Chan School of Public Health Professional Development Support Fund	Oct 2023
Outstanding Undergraduate Thesis, Sichuan University	Nov 2022
Outstanding Undergraduate Leader, Sichuan University	Oct 2021
The Comprehensive Scholarship of Sichuan University for Dual Degree Students	Mar 2021
Silver Award, 2020 “Internet +” College Students Innovation and Entrepreneurship Contest	Dec 2020
The Comprehensive Scholarship of Sichuan University for Undergraduates	Oct 2020

PEER REVIEW ACTIVITIES

Archives of Public Health	1 manuscript
BMC Health Services Research	3 manuscripts
BMC Public Health	1 manuscript
Epidemiologic Reviews	1 manuscript
Health Economics Review	1 manuscript
Health and Quality of Life Outcomes	1 manuscript
International Journal for Equity in Health	1 manuscript
The European Journal of Health Economics	1 manuscript
ISPOR Asia Pacific Summit 2026	abstract peer reviewer

LEADERSHIP

Global Health and Wellness Research Association of Sichuan University
President of the Association Sep 2020 – Jun 2021

- Organized six expert lectures on statistical package applications and academic frontiers.
- Assisted members in submitting Student-Led Research and Innovation Programs to Sichuan University on behalf of the association.